

Untangling the oral–gut axis in the pathogenesis of intestinal inflammation

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Abstract

An increasing body of literature reveals that host–microbe networks are well coordinated and impact human health and disease. Recently, it has become evident that an abnormal alteration in bacterial configuration in the oral cavity, namely oral dysbiosis, caused by periodontal inflammation, is associated with various distant inflammatory diseases, including inflammatory bowel disease. However, the extent to which the relationships between oral and distant disorders are merely an association or are causally triggered by oral microorganisms remains debated. In this mini-review, we highlight mechanisms in inter-related organ system diseases, particularly the one between oral and gut inflammation. Further, we discuss clinical perspectives and propose a novel concept of a multi-hit hypothesis in the pathogenesis of gut inflammation, on the basis of our updated knowledge of shared microbiological and immunological pathways between the oral and gut mucosae.

Keywords: inflammatory bowel disease, intermucosal interactions, oral bacteria, periodontitis, systemic organ interactions

Introduction

The human body is home to 3.8 trillion microorganisms, almost equivalent to the number of human cells and collectively referred to as the human microbiota (1, 2). Because of environmental differences, each site in the body has a distinct microbial ecosystem. Among them, the gut microbiota encompasses the richest and most diverse microbial community, which contributes to host physiologic development and maintenance, including nutrient digestion, education of the host immunity and defense against colonization by pathogenic microorganisms (3–5).

Inflammatory bowel disease (IBD) is represented mainly by two subtypes of idiopathic intestinal disorders, Crohn's disease (CD) and ulcerative colitis (UC), that cause prolonged inflammation of the intestinal tract. In the pathogenesis of IBD, it has been postulated that immune-mediated pathologies arising from dysregulated immune responses against intestinal microorganisms play a critical role (6). However, the questions of which microorganisms contribute to the pathogenesis and how they do this remain unanswered. In this context, several lines of evidence show that the prevalence of periodontitis is consistently shown to be higher in patients with IBD than in healthy individuals (7–12). Also, a significant enrichment of oral bacteria in the gut (e.g. *Fusobacteriaceae*, *Veillonellaceae*, *Pasteurellaceae*, *Enterobacteriaceae*, *Nisseriaceae*, *Gemellaceae*), including

pathobionts associated with periodontitis, has also been shown (13–16). However, the causal link between periodontal and gut inflammation has been unexplored. In this regard, our recent studies uncovered the causative and complex interplay between the oral and gut mucosae in the pathogenesis of gut inflammation.

In this review, we summarize the current knowledge of the oral–gut axis during gut inflammation and the clinical perspectives and propose a multi-hit hypothesis focusing on the oral–gut axis in the pathogenesis of gut inflammation.

Gut colonization by oral pathobionts aggravates gut inflammation (bacteria-mediated direct pathway)

Research assessing the impact of direct gut colonization of human-derived oral bacterial strains (e.g. *Porphyromonas gingivalis*, *Fusobacterium nucleatum*) on murine intestinal inflammation has dramatically expanded over the last two decades (17). However, because of differences between human and murine immune systems (18, 19) and microbial composition (20), the species-mismatch model has limitations, particularly for addressing the causality with precise host–microbe interactions during disease development within a single living animal. Thus, despite a wide variety of association-based studies, the causal link between

periodontal and intestinal pathology has been unanswered for a long time.

To circumvent this limitation, our group recently conducted a series of species-matched experiments in mice and discovered the causal link between the mouth and gut during the pathogenesis of intestinal inflammation (21). In this study, by combining ligature-induced murine periodontitis and dextran sodium sulfate (DSS)-induced acute colitis models, we confirmed that mice with periodontitis exhibited more severe colitis than those without periodontitis. Further, we observed that the expansion of oral pathobionts, particularly Enterobacteriaceae (e.g. *Klebsiella* spp. and *Enterobacter* spp.), and subsequent gut colonization by those pathobionts are critical for the exacerbation of colitis. Moreover, the colitogenic capacity of those oral pathobionts has been confirmed in the gut of genetically susceptible IL-10^{-/-} mice (but not wild-type B6 mice). Mechanistically, we also revealed that oral pathobionts were capable of inducing colonic IL-1β production by activating the inflammasome pathway in intestinal macrophages in the inflamed gut, thereby aggravating the

intestinal pathology (21). Taking advantage of a species-matched model, we further addressed the essential questions of which factors determine the efficient gut colonization by oral pathobionts. No overt enrichment of oral pathobionts was observed in the healthy gut even in the presence of periodontitis, whereas a significant level of gut colonization by oral pathobionts was detected in mice with both oral and gut inflammation.

Although more studies will be required to clarify the detailed mechanisms, on the basis of our findings and previous studies describing the physiological barrier against ingested bacteria, we herein propose a novel concept of the ‘multiple-hit’ hypothesis for the successful translocation of oral pathobionts to the gut during the development of oral bacteria-driven intestinal inflammation as follows (Fig. 1).

The dysbiotic oral cavity as a reservoir of oral pathobionts

Within the oral ecosystem, niches created by disease conditions such as periodontitis shape microbial communities

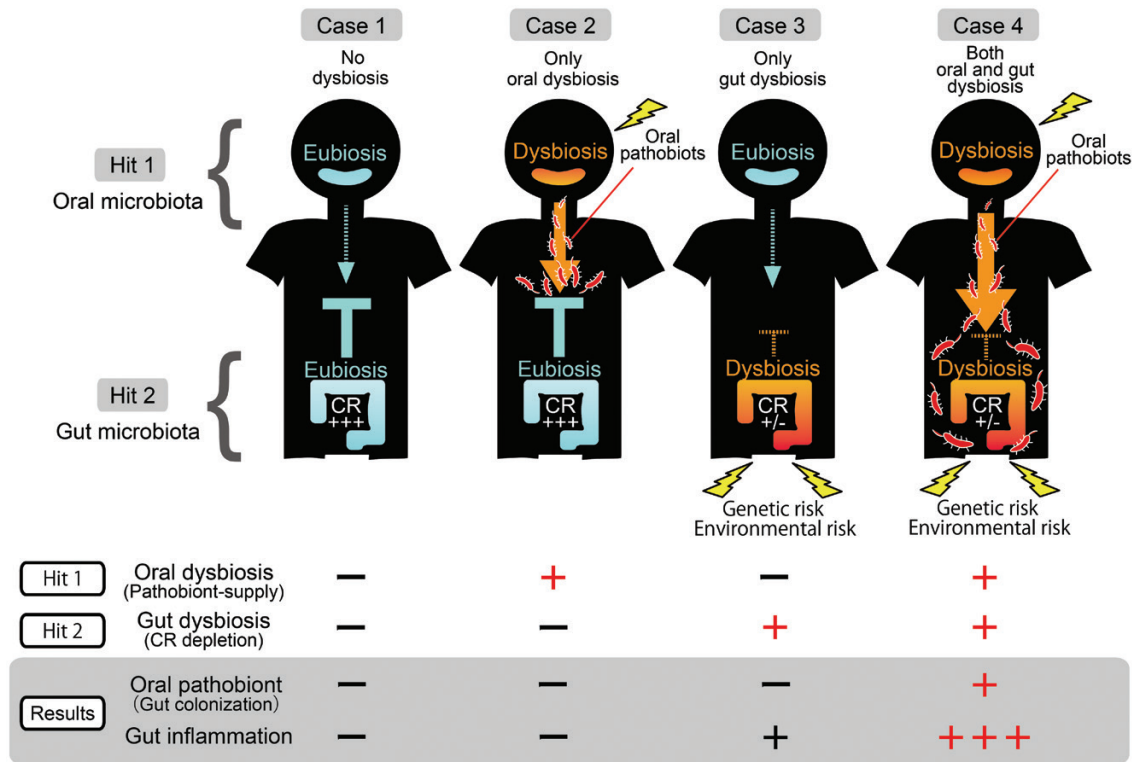


Fig. 1. A multi-hit hypothesis for the successful gut colonization by oral pathobionts in the pathogenesis of gut inflammation. The development of oral bacteria-associated gut pathology can be defined by the following conditions. (Case 1) Without any complications in both the oral and gut compartments with eubiotic microbial compositions, neither the expansion nor dislocation of oral pathobionts to the gut occurs, thereby keeping the host healthy and in a state of homeostasis. (Case 2) The presence of periodontal disease causes oral dysbiosis that allows the oral pathobionts with colitogenic potential to expand in the oral cavity. However, those pathobionts fail to reach the gut as long as intact barrier functions (e.g. CR) along the GI tract are present. (Case 3) Similarly, only having preceding barrier dysfunction along the GI tract, such as gut dysbiosis, is insufficient to make the host susceptible to gut colonization by oral pathobionts because the burden of oral pathobionts in the oral cavity is not enough to enhance the gut pathology. (Case 4) Concurrent presence of oral dysbiosis (sufficient supply of oral pathobionts) and gut dysbiosis (barrier dysfunction against ingested microorganisms) are necessary and sufficient conditions for the successful gut colonization by oral pathobionts, thereby aggravating intestinal inflammation. In addition, the physiologic impact based on the proposed multi-hit hypothesis can be amplified by the involvement of other factors facilitating gut colonization by oral pathobionts. In parallel, successful gut colonization by oral pathobionts also serves as a trigger for the expansion of orally primed T cells in the gut and engagement of microbiological and immunological pathways along the mouth–gut axis during the pathogenesis of the intestinal inflammation.

and foster certain pathogenic oral bacteria. For example, perhaps the best-studied oral pathobiont is *P. gingivalis*, which expands in the oral cavity of patients with periodontitis and is extensively reported to play a role in other disorders including atherosclerosis, pneumonia, rheumatoid arthritis and Alzheimer's disease (22–26). In this context, we identified that oral dysbiosis generated by periodontal inflammation is a first hit (prerequisite) that is responsible for the drastic expansion of colitogenic oral pathobionts in the oral cavity. It is followed by an increased chance of successful transmission of those pathobionts of oral origin to the intestine (Fig. 1) (21). This is supported by our data delineating the concurrent expansion of oral pathobionts (e.g. *Klebsiella* and *Enterobacter* spp.) in both oral and gut compartments in mice with ligature and DSS treatment, whereas there was no sign of enrichment of oral pathobionts in the gut of non-ligated animals, even in the presence of DSS treatment (Fig. 1; Case 3 and Case 4). In addition, given that up to 25% of IBD cases develop during childhood or adolescence (27), factors in the oral conditions other than periodontal inflammation may also impact the expansion of certain oral pathobionts. In this regard, a recent study found that fixed orthodontic appliances in school-age children (median age, 12.7 years old for males and 13 years old for females) are associated with a higher incidence of oral colonization with members of the Enterobacteriaceae family, including *Klebsiella* and *Enterobacter* spp. (28). Although the clinical association between IBD and oral conditions besides periodontitis, including orthodontics, remains unknown, considering the colitogenic capacity of human and murine Enterobacteriaceae (21, 29), this may indicate a possible risk of the exacerbation of IBD. Interestingly, the authors also found that children with habitual chronic nail-biting and undergoing orthodontic treatment exhibited the highest colony-forming units (CFUs). This may also imply that not only intrinsic expansion of oral pathobionts (e.g. inflammation, orthodontics) but also extrinsic factors (e.g. nail-biting) allow the opportunistic environmental microorganisms including Enterobacteriaceae to colonize the oral cavity and may increase the risk of the development of gut inflammation caused by orally derived pathobionts (28).

Disruption of gut colonization resistance

The colonization resistance (CR) conferred by a harmonious microbial configuration in the healthy gut is considered critical for preventing ectopic colonization by ingested oral bacteria (30–32). Recent studies have demonstrated that multiple factors (e.g. antibiotics, diets, artificial sweeteners) that induce gut dysbiosis also cause impaired gut CR, which contributes to the efficient enrichment of oral bacteria in gut diseases including IBD (15, 32). Thus, as the second hit, impaired gut CR induced by gut dysbiosis is required for efficient gut colonization by oral pathobionts that successfully pass through the gastric barrier (Fig. 1). This notion is supported by our data from animal studies showing that mice without preceding gut dysbiosis (resulting from DSS-induced inflammation) do not exhibit enrichment of oral pathobionts in the gut, even in the presence of periodontitis (Fig. 1; Case 2 and Case 4) (21). Consistent with findings in animals (21), recent research by our group using matched saliva and stools from IBD patients with or without periodontitis revealed that

the gut microbial configuration in patients with IBD was significantly more similar to their oral microbiome than the oral and gut microbiomes were in healthy controls. This suggests that preceding impairment of gut colonization is another prerequisite for gut colonization by oral bacteria in IBD patients (33). Importantly, although further comprehensive validation studies with larger sample size and patient heterogeneity will be needed, we observed a potential adverse impact (e.g. increased disease activity) of early periodontitis on intestinal inflammation in patients with CD in the study (33). Supporting this idea, other groups also showed that ectopic colonization of specific oral microbes such as *Streptococcus salivarius* is enriched more in CD subjects with a higher Crohn's Disease Activity Index (184.3 ± 2.9 versus 67.1 ± 82.5 , $P = 0.012$) and active disease status (diarrhea/abdominal pain/blood in stool/fever and fatigue, $P = 0.016$) (34). Consistent with animal findings, our study also demonstrated that the Enterobacteriaceae family was significantly enriched within patients who had CD with periodontitis compared with those who did not have periodontitis. However, in our study, we could not identify specific human oral pathobionts associated with IBD because of the insufficient sample number and/or selection bias, as the participants in this study were relatively younger and therefore had developed only mild (i.e. incipient) periodontitis. Clearly, further studies will be required to clarify the extent to which the amassed bacteria harbor colitogenic capacity in humans.

Other factors facilitating gut colonization by oral pathobionts

Although the dissemination routes of oral pathobionts to the gut remain uncertain, given the fact that the intestinal tract is anatomically connected to the mouth and continuously exposed to ingested foods as well as saliva [~ 1.5 L daily containing 1.5×10^{12} oral bacteria (2, 35)], it is plausible that oral pathobionts can reach the gut by passing through the enteral side of the gastrointestinal (GI) tract. In this context, in addition to gut CR conferred by the commensal gut microbiota, impaired barrier function along the GI tract may affect the efficiency of gut colonization by oral pathobionts. One example is the attenuated gastric acidity in the stomach that deters the successful transmission of ingested bacteria to the lower GI tract (36, 37). Previous studies demonstrated that more than 99% of swallowed microbes of oral origin are inactivated as they pass through the stomach (2, 38).

In fact, gut colonization by oral bacteria (e.g. *Streptococcus* spp., *Veillonella* spp.) is observed in patients who have gastric achlorhydria caused by the long-term use of proton pump inhibitors (PPIs) (39). Consistently, patients with gastroesophageal reflux disease (GERD) treated with long-term PPI therapy also exhibit higher oral bacterial accumulation in the gut than healthy individuals (29). With respect to the clinical impact of PPIs on the pathogenesis of IBD, previous studies have demonstrated that the PPI use could be a risk factor for the adverse outcomes in IBD (40, 41). Importantly, the gastric pH of patients with IBD (CD pH 2.4, range 1.5–4.1; UC 1.95, range 1.55–4.4) is significantly higher than that of healthy individuals (pH 1.55, range 0.95–2.6) (42). Further, given that the reported murine stomach pH is 3–4 (43), murine models might be considered to mimic the attenuated gastric acidity observed in IBD patients rather than

that of healthy individuals (21). However, despite the higher gastric pH in mice, we observed no colonization with or less enrichment of oral pathobionts in mice with periodontitis compared with those that had both periodontitis and colitis, implying that attenuated gastric acidity *per se* is insufficient to allow the successful gut colonization by oral pathobionts (21). Although additional studies are required to clarify the importance of gastric acidity in the pathogenesis of IBD, these studies imply that attenuated gastric acidity in the stomach is conducive to the efficient transmission of ingested bacteria to the lower GI tract.

Likewise, in addition to their immune modulatory functions, bile acids have an amphipathic detergent activity that disrupts bacterial cell walls and can serve as potent antimicrobial agents, thereby functioning as a physiological barrier against harmful invasion by microorganisms (44–46). In this context, a previous study delineated that patients with IBD exhibit an alteration in their bile acid profile and gut microbial composition. Particularly, the concentration of secondary bile acids [e.g. lithocholic acid (LCA), deoxycholic acid (DCA)] was profoundly decreased, whereas the proportion of primary and conjugated bile acids was increased in patients with IBD (47). Importantly, although the detailed mechanisms remain unclear, this elevation of the primary and conjugated bile acids [e.g. cholic acid (CA), taurocholic acid (TCA) and glycochenodeoxycholic acid (GCDCA) or taurochenodeoxycholic acid (TCDCA)] was positively correlated with the increased abundance of certain types of oral pathobionts, including *Klebsiella* spp. (21, 29, 48), implying the potential importance of bile acids in controlling gut colonization by oral pathobionts.

Translocation of orally primed immune cells to the gut enhances gut inflammation (immune cell-mediated indirect pathway)

Accumulated evidence indicates that periodontitis not only forces a focal inflammation in periodontal tissues but can, in turn, affect the pathogenesis of systemic diseases (26). In fact, patients with periodontitis and IBD have also been linked by shared immunological traits, including the generation of certain types of innate (e.g. IL-6, IL-1 β , myeloid cells) and adaptive (e.g. Th1, Th17) immune responses (16). Given a significant correlation between gingival and intestinal inflammation scores (49), these findings indicate an immunological link between periodontal and gut inflammation.

In this context, we recently uncovered that the unfavorable immunological cross-talk between the mouth and the gut contributes to the pathogenesis of gut inflammation. As previously mentioned, in our study, ligature-induced murine periodontitis increases the susceptibility to DSS-induced colitis through direct gut colonization by oral pathobionts (Fig. 1) (21). In parallel, even though we employed the acute DSS-induced colitis model, we unexpectedly found a significant increase in Th17 and Th1 cells in the colonic mucosa of ligature–DSS mice compared with mice only treated with DSS. Because of the importance of Th17 cells in periodontitis (50) and the transmission capability of orally derived leukocytes in the oral draining lymph nodes to the systemic organs, including the gut (51), we hypothesized that pathogenic T

cells found in the acute DSS-induced colitis model could be of oral origin.

To test this, we conducted a series of experiments and obtained the following results (21):

1. Effector memory T (T_{EM} ; CD3⁺CD4⁺CD44^{hi}CD62L^{lo}) cell numbers were significantly increased in the cervical lymph nodes (cLNs) of mice with periodontitis compared to mice that did not have periodontitis.
2. Consistent with the previous reports (52, 53), those orally primed T cells generated in mice with periodontitis had an IL-17A-producing retinoic acid-related orphan receptor γ t (ROR γ t)⁺ Th17 phenotype and were capable of responding to oral pathobionts (e.g. *Klebsiella* spp. and *Enterobacter* spp.) that expand in the inflamed oral cavity.
3. Furthermore, gut-homing markers such as α 4 β 7 integrin and C–C chemokine receptor 9 (CCR9) were expressed on these oral Th17 cells, indicating their gut tropism.
4. Next, the transmigration of orally primed Th17 cells to the gut mucosa was confirmed by employing the cellular trafficking model mouse strain expressing the Kaede protein (51).
5. Moreover, orally primed Th17 cells were colitogenic only in the presence of *Klebsiella aerogenes*, which are oral pathobionts isolated from ligatured animals.
6. Of note, administration of an IL-1 receptor antagonist (anakinra) ameliorated the severity of colitis caused by the transmigration of orally primed Th17 cells. Given the Th1-skewing properties of IL-1 β , intestinal IL-1 β production from gut colonization by oral pathobionts (e.g. *K. aerogenes*, Fig. 1) not only serves as a pro-inflammatory cytokine but also acts as a Th1-skewing factor for generating Th1/Th17 cells (ROR γ t⁺ T-bet⁺) associated with IFN- γ production, which are considered more pathogenic than Th17 cells in the development of IBD (54–58).

Perspectives

It has become evident that there is a causal link between periodontal and intestinal pathology through complex microbiological and immunological interactions (21). However, there is still a knowledge gap. For the microbiological pathway, these unresolved questions include: other than oral inflammation, what other conditions allow the oral pathobionts to expand in the oral cavity? Also, which factors and combinations of barrier malfunctions (e.g. gastric acidity, CR, others) are responsible for the ectopic gut colonization by oral pathobionts in humans? In addition, the microbial dissimilarity between humans and mice has been shown (20). Although the colitogenic oral pathobionts (e.g. *K. aerogenes*) that we identified in murine studies are genetically similar to *K. aeromobilis*, reported to be a colitogenic oral pathobiont in patients with IBD, it remains uncertain whether current findings in the oral–gut axis are readily translatable to humans. In this context, mucosa-associated bacteria but not luminal bacteria are reported to be strong Th17 inducers by attaching to epithelial cells in the gut (59–61), indicating a functional deviation between bacterial communities in the lumen and mucosa. Considering the biogeographical difference between luminal and mucosa-associated gut microbiota

(62, 63), sampling methods enabling the recovery of mucosa-associated bacteria (e.g. endoscopic mucosal biopsy, endoscopic brush samples) might be important in the context of human studies. At this point, the study of the oral–gut axis in the IBD is still in its infancy and no clinical interventions targeting this particular axis are available on the market. However, future investigations of the microbiological and immunological pathways that link the mouth and gut will pave the way for developing novel diagnostic and therapeutics for IBD (17). In fact, a recent study supports the idea of targeting immunological pathways in the context of periodontitis-driven aggravation of intestinal inflammation. In this study, using the ligature–DSS murine model, the authors demonstrated that a reduction in the Th17 cell population in inflamed periodontal tissues upon ligature treatment by the local injection of certain types of exosomes is capable of alleviating the severity of experimental colitis in mice with both ligature and DSS treatment, compared with DSS treatment-only mice (64).

Although the detailed mechanisms remain unexplored, this highlights the clinical potential of therapies targeting the oral–gut axis during the development of IBD. Moreover, given the recent study highlighting the adverse impact of obesity-associated gut dysbiosis on the pathogenesis of periodontitis through the elevation of uric acid (65), a better understanding of bidirectional interactions between the mouth and gut constitutes an essential step in developing better diagnostics and therapeutics for both periodontal and intestinal pathologies. To embrace these challenges, we would benefit from multidisciplinary collaborations among immunologists, microbiologists, physiologists and clinicians, together with bioinformaticians for integrating large datasets acquired from both organs as well as with microbial ecologists for their theoretical frameworks to understand such extraordinarily complex interactions.

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